



Team 15 Sprint 2 Planning Document

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Sprint Overview

This sprint is building on the foundation that we created in Sprint 1, integrating AI into the features that we have created, and creating fully AI features to allow for assistance and advice within workouts and dieting. We are already able to create accounts and manage our workouts within the app, and the goal of this sprint is to refine these processes further, while adding artificial intelligence that will bring a unique aspect to our app. Users will have suggestions for workouts and diets tailored to their needs, which will make it much easier for them to get used to the gym.

SCRUM Master: Drew Balaji

Meeting Plan: 3:00 to 4:00 PM Thursday and Sunday at Hicks Undergraduate Library

Risks and Challenges:

This sprint is fundamentally based on implementing AI-focused features and integrating AI into parts of our project. Because we are not training our own AI, we will have to integrate existing models, such as GPT models, into our code and have it perform the functions that we want properly. Most, if not all, of us have not integrated AI into a project of ours, so this brings a completely new element into our code that we will have to figure out. Along with that, ensuring that our AI works all the time and that it does not crash and fail based on updates or changes in models is a very important challenge for us to face.

Current Sprint Detail

User Story 22

As a gym user, I will be able to generate an all-inclusive, tailored, generic Quick Workout, accounting for every possible profile characteristic.

Task	Description	Estimated Time	Member Assigned
1	Profile Retrieval: FastAPI endpoint that fetches health goals, health status, bio, workout history, diet history, and formats into a structured string ready to be used as LLM context	3 Hours	Luke
2	Workout Embedding: Generate embeddings for each workout (Workouts/Exercises/workout_exercises) using an embedding model (ChatGPT/Claude)	3 Hours	Roan
3	Meal Embedding: Generate embeddings for both restaurant menu meals and recipes (Meals/Ingredients/meal_ingredients)	3 Hours	Roan
4	Pinecone Ingestion: Upsert workout embeddings (2) into a Pinecone index with tags as metadata. Upsert meal embeddings (3) into a Pinecone index with tags as metadata	4 Hours	Pratik
5	Profile Embedding: Generate embeddings from structured profile retrieval strings (1)	2.5 Hours	Luke
6	Pinecone Query: Query profile embeddings (5) to Pinecone (4) then retrieve the most relevant workouts for them.	4 Hours	Pratik
7	Prompt Construction: Using Pinecone query (6) results and profile characteristics (1), generate "I am 22 years old, I exercised xyz this week,, generate a workout for me."	3 Hours	Pratik

8	LLM Integration: FastAPI endpoint that chains steps together, calls the LLM API, and returns structured workout recommendations.	3 Hours	Pratik
9	Frontend Integration: UI button, UI displays, api.ts routes	2 Hours	Artem
10	Testing to ensure proper completion of tasks for further implementation of user story tasks, along with ensuring a proper workout is generated for specific characteristics.	2 Hours	Pratik

Acceptance Criteria:

- User presses the Quick Workout button, then sees a display of a generated workout with exercises
- The user can manually make any changes they desire
- User presses Start Workout to start the timer and add to the log, or Nevermind to back out

User Story 23

As a gym user, I will be able to generate a tailored workout purely based on a specific muscle group.

Task	Description	Estimated Time	Member Assigned
1	Develop a prompt to LLM based on the muscle group selected by a user, that will return a different workout based on the muscle group input	2.5 Hours	Luke
2	UI for selecting a muscle group to workout, or searching/choosing a muscle group from a static list in the database	3 Hours	Luke
3	Testing to ensure that a proper workout is generated based on the given muscle group	1.5 Hours	Luke

Acceptance Criteria:

- If the LLM-app integration is properly executed, if a muscle group for a workout to be generated for is selected, a proper workout should be produced by the LLM and displayed to the user.
- If the muscle group-centered prompting works, if a user selects a specific muscle group to work out, the user should be given a workout that properly corresponds to the given muscle group.

- If the error handling is done properly, if the user generates a workout for a muscle group without explicitly saying a muscle group, the user should be prompted with an error message, and no workout should be generated.

User Story 24

As a gym user, I will be able to generate an all-inclusive, tailored, generic Quick Meal, accounting for every possible profile characteristic.

Task	Description	Estimated Time	Member Assigned
1	Pinecone Query: Query profile embeddings to Pinecone then retrieve the most relevant meals for them.	3 Hours	Pratik
2	Prompt Construction: Using Pinecone query results and profile characteristics, generate "I am 22 years old, I ate xyz this week, ..., generate a meal for me."	4 Hours	Pratik
3	LLM Integration: FastAPI endpoint that chains steps together, calls the LLM API, and returns structured meal recommendations.	3 Hours	Artem
4	Frontend Integration: UI button, UI displays, api.ts routes	2 Hours	Artem
5	Testing button functionality, and making sure that all possible characteristics work for generating a meal.	1.5 Hours	Artem

Acceptance Criteria:

- User presses the Quick Meal button, then sees a display of a generated recipe containing ingredients, macros, steps to make the recipe, and how it impacts your set goals.
- If the User doesn't want to cook, they can press a button to see meals of similar criteria at national chain restaurants
- User can add the at-home or menu meal to their log

User Story 25

As a dieter, I will be able to chart intake and set goals for macro and water.

Number	Description	Estimated Time	Member Assigned
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1	Implement logic to allow users to set a goal for charting	3.5 Hours	Drew
2	Implement logic to let users set different goals through the use a scroll wheel	3.5 Hours	Drew
3	Implement routing to let the meal functionality and goal trackers communicate with one another	5 Hours	Drew
4	Implement a UI to show the goal and show user progress through the use bar charts	6.5 Hours	Drew
5	Create user-facing tests to ensure that the logic and UI work	1.5 Hours	Drew

Acceptance Criteria:

- Users would be able to set initial goals and add amounts during the day
- Users will be able to change their goals after setting the amounts if they need to
- Users will be able to see their goals automatically populate with added meals

User Story 26

As a new user, I will have my day-one caloric goals displayed based on onboarding responses.

Task	Description	Estimated Time	Member Assigned
1	Add daily calorie goal column to profile in database, both schema and inserting pipeline	1 Hour	Roan
2	Process user onboarding responses to return a recommended day-one calorie target from metabolic rate equation	2 Hours	Roan
3	Server capabilities added onto the end of the existing onboarding endpoint	2 Hours	Roan
4	Add a calorie widget on the diet UI tab, routed front → back	2 Hours	Luke
5	Test: navigate to the diet tab and see today's calorie goals displayed	1 Hour	Luke

Acceptance Criteria

- User successfully submits an onboarding quiz, finalizing a new account creation
- The user can navigate to the diet UI tab and see their calorie goals displayed in the corner
- User can close the app, reopen it, and see the same number displayed

User Story 27

As a dieter, I will press a button to recalibrate my caloric goals based on recorded behavior.

Task	Description	Estimated Time	Member Assigned
1	Update at-home & menu meal logging capabilities to include time and calories - frontend, database, and endpoint functions	4 Hours	Roan
2	Develop queries that return goal, consumed, and remaining calories to endpoints	2 Hours	Roan
3	Extend profile retrieval and embeddings to include daily calorie goal, recent calorie consumption, and workout frequency	2 Hours	Roan
4	Prompt Construction: Leveraging workout + diet pinecone query results, draft "this user's goal is x and they've done yz, should their calorie goal be adjusted?"	2 Hours	Roan
5	LLM Integration: combines retrieval, embeddings, queries, and prompts to call the LLM and updating server function	4 Hours	Roan
6	Update calorie widget on the diet UI tab with a recalibrate button, consumed & remaining calories, routed front → back	2.5 Hours	Pratik
7	Test: store current calorie goal, log a new workout or meal, call LLM and update calorie goal, check if different than current	1.5 Hours	Artem

Acceptance Criteria

- User acknowledges the current status, then proceeds to log a workout or meal
- User returns and presses the recalibrate button; adjustments are displayed
- User switches to another tab and returns to see the same newly adjusted calorie goal

User Story 28

As a user, I would be able to assign macros and ingredients to my meals when I add them.

Number	Description	Estimated Time	Member Assigned
1	Implement logic to allow users to add the different macro amounts (protein, fat,	3 Hours	Drew

	sugar, etc.)		
2	Implement logic to allow users to assign different ingredients to a meal	3 Hours	Drew
3	Create a dropdown UI to nest the macro and ingredient assigner to the meal adding screen	4 Hours	Drew
4	Create user-facing tests to ensure that the logic and UI work	1 Hour	Drew

Acceptance Criteria:

- Users will be able to see different macros to write into each meal card and see an empty list to add ingredients
- Users will be prompted to add macros or ingredients when left blank
- Users will be able to edit macros and ingredients after saving their meals

User Story 29

As a user, I would like to be able to have weekly and monthly diet and workout reports generated.

Task	Description	Estimated Time	Member Assigned
1	Reuse Tasks 1-5 from User Story 22 Profile Retrieval, Workout Embedding, Meal Embedding, Pinecone Ingestion, Profile Embedding	1.5 Hours	Luke
2	Determine statistics/values to return for each report	2 Hours	Luke
3	Develop a query/prompt to generative AI to calculate statistics and develop advice/encouragement for next week/month	1.5 Hours	Luke
4	Parse data returned from LLM, and retrieve other statistics/values from database for display to users.	2 Hours	Luke
5	Tests to ensure that the proper data is retrieved/returned based on a template user and prompt	1.5 Hours	Luke
6	UI Formatting to display information retrieved from the database/LLM	2.5 Hours	Luke

7	Add temporary UI button to generate a weekly/monthly report based on sample week/month data on command (for presentation)	1.5 Hours	Luke
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Acceptance Criteria:

- Given that the reports are generated properly, if the end of a week or month occurs, the user should have a report generated with a value for each statistic/value listed.
- Given that the UI is implemented as intended, if the end of a week or month occurs, the user should be able to see all statistics/values in a neat UI screen.
- Given that the reports are generated properly, if a report is generated for a specific user, the report should reflect the data for that user specifically, and not any other user in the database.
- Given that the UI is connected properly, when a report is generated and displayed, it should be a pop-up that, when closed, should return to the home screen in the app without any error.

User Story 30

As a gym user, I can have my weight, sets, and reps tailored for a specific ongoing exercise, accounting for all necessary logged progress

Task	Description	Estimated Time	Member Assigned
1	Create a UI button that allows users to prompt an LLM for questions	2 Hours	Jack
2	Create a UI button that allows users to prompt an LLM for questions	1 Hour	Jack
3	Retrieve an embedding of the LLM's response	2 Hours	Pratik
4	Generate a UI to display the LLM's response and information	2 Hours	Jack
5	Testing to ensure that for a given exercise, the total number of reps and sets makes sense and is optimal for the specific user.	1 Hour	Artem

Acceptance Criteria:

- Given that the UI button functions properly, if the user presses it, they should be provided with a place to input an exercise to have weight, sets, and reps tailored to.
- Given that the LLM prompting works, if a user selects a specific exercise, they should be given sets, reps, and weight that are reasonable for that exercise

- Given that the UI display is correct, when the user generates weight, sets, and reps for an exercise, the output of the LLM should be displayed for the user to see in their workout.

User Story 31

As a gym user, I will have access to a help feature that suggests alternative options if a machine or exercise isn't available or is undesirable.

Task	Description	Estimated Time	Member Assigned
1	Create an associative button to append to exercises for every workout	2 Hours	Jack
2	Connect button to LLM to prompt for an associated exercise or machine	1 Hour	Jack
3	Generate a UI to display the LLM's recommended alternatives upon user clicking the alternative button	3 Hours	Jack
4	Testing to make sure that a similar machine or exercise is suggested if the user can not do a specific exercise/machine.	1 Hour	Artem

Acceptance Criteria:

- Assuming that the associative button works properly, if a user does not want to do a certain exercise, the user should be able to press a button to switch the exercise.
- Assuming the LLM to button connection is correct, whenever the user presses a button to switch an exercise, the LLM should be prompted with the initial exercise and find a unique, new exercise that fits the same needs.
- Assuming the LLM prompting works, if a user continuously attempts to get new exercises, the user should be prompted with a message forbidding them to do that after 5 attempts

User Story 32

As a beginner in my fitness journey, I will have access to a help feature that provides assistance on exercises.

Task	Description	Estimated Time	Member Assigned
1	Implement a help button/icon that users can select to access exercise assistance	1 Hour	Artem
2	Implement logic to retrieve exercise help information (detailed step-by-step instructions on how to complete an	4 Hours	Artem

	exercise, exercise tips, links to videos demonstrating the exercise) from the backend.		
3	Create a UI component to display exercise help information.	2 Hours	Artem
4	Create tests to ensure that users are able to get help for exercises, and the help makes sense for each exercise.	1.5 Hours	Artem

Acceptance Criteria:

- When a user is viewing an exercise, there will be a button they can press to access exercise assistance.
- When the user presses the help button, help information will be displayed.
- When the user closes the help feature, they will be able to return to the exercise screen.

User Story 33

As a gym user, I would like to view a graph of weight progression for a specific exercise

Task	Description	Estimated Time	Member Assigned
1	Create an embedded button for exercises to allow users to view a weight progression graph	2 Hours	Jack
2	Prompt the LLM for data points related to weight progression to generate for the user	2 Hours	Jack
3	Generate and display a graph showing the users weight progression and future weight progression	3.5 Hours	Jack
4	Testing that the graph is populated correctly, and it is accurate for the information that is provided.	1 Hour	Pratik

Acceptance Criteria:

- If the button is implemented, when the user clicks on the button, they are transported to a graph where they can see their exercise weight progression
- If the graph is created properly, when the user has a graph generated, it should be representative of the workouts the user has completed.
- If the graph is created properly, when the user sees the graph, they should be able to see the weight they started at and where they are based on their last workout.

Sprint 1 - Leftover User Stories and Tasks

User Story 11

As a dieter and athlete, I would be able to switch between imperial and metric systems to measure weights in the gym and portions in the kitchen.

Number	Description	Estimated Time	Member Assigned
1	UI switch to switch between measurement systems while the task is being completed	1.5 Hours	Luke
2	Formula to calculate conversions, generalized per use case (gym/kitchen), either manually or through LLM	1 Hour	Luke
3	For all functionality that involves weights, update all values with the proper conversions and proper units based on the setting in the database	2.5 Hours	Luke
4	Tests: button works, accurate conversions for a recipe and a workout	1 Hour	Luke

Acceptance Criteria:

- Given that metric-to-imperial conversion is completed in the app, when the user selects the option to switch systems, they should be able to switch back and forth between the two.
- Displayed conversions between the metric and imperial measuring systems are accurate.
- The user can complete tailored exercises or recipes based on the conversion processed for them, depending on their settings.

User Story 14

As a user, I would like to be able to select a location, so I can see gyms and people near me.

Number	Description	Estimated Time	Member Assigned
1	Create a UI button to open the select menu	1.5 Hours	Jack
2	Synchronize with Maps API to embed a display of nearby gym locations	3 Hours	Jack
3	UI to request user permissions for location access	1 Hour	Jack
4	Query the database to connect with	3 Hours	Jack

	nearby friends that also have location access		
5	Create tests to ensure proper location syncing in and finding the proper gyms	1.5 Hours	Jack

Acceptance Criteria:

- Given that the UI button is implemented correctly, the nearby gyms menu should display to the user on click.
- Given that the UI for requesting user permissions is implemented correctly, the user should receive a pop-up for location access upon attempting to view nearby gyms.
- Given that the Maps API is properly implemented, the user should be able to view an embedded map with nearby gyms pinned.
- Given that the Maps API and database are properly implemented, the user should be able to view nearby friends and users.

User Story 15

As a user, I would like to be able to enable certain accessibility features (dark mode, large text).

Number	Description	Estimated Time	Member Assigned
1	Implement persistent storage of accessibility settings (local storage or backend)	1 Hours	Artem
2	Load and apply accessibility settings automatically when the application starts	2 Hours	Artem
3	Test to ensure that accessibility features work properly, and load to the correct state when opening the app.	1 Hour	Artem

Acceptance Criteria:

- Users who have used accessibility settings to make the app easier for them to use will not have to reapply those settings after every app restart.

User Story 18

As a dieter, I would be able to view the nutritional facts of different meals from national restaurants.

Number	Description	Estimated Time	Member Assigned
1	Design schema to manage restaurant meals	1 hour	Roan
2	Implement schema and populate table in database	1 hour	Roan
3	Design query to grab meals based on different contexts	3 hours	Roan
4	Display meals that meet declared criteria	2 Hours	Roan
4	Test: high protein meals appear for high protein criteria, Chipotle restaurant shows Chipotle meals	1 Hours	Roan

Acceptance Criteria:

- The user can filter menu meals by restaurant
- The user can filter menu meals by protein source
- The user can filter menu meals by nutrition margin (protein > 30)

User Story 19

As a user, I will be able to bypass manually inputting username and password during login with a Remember Me feature

Task	Description	Estimated Time	Member Assigned
1	Design server side logic, database schema, and access code hash functions	4 hours	Roan
2	Create and route with frontend UI capabilities	2 hours	Roan
3	Debug tasks 1 and 2	2 Hours	Roan
4	Test: open the app and login, close the app, reopen the app and see if account is remembered	1 Hours	Roan

Acceptance Criteria:

- The user can seamlessly open the app and bypass login during times of frequent back and forth from the phone, like while working out
- User doesn't waste time on repeated, tedious tasks that are required to use the service
- The feature only applies to certain criteria and will expire for users after 30 days

User Story 20

As a user, I would like to be able to read and agree to the Terms and Conditions of the service.

Number	Description	Estimated Time	Member Assigned
1	Draft the terms and conditions policy for users to read and accept	2.5 hours	Jack
2	Frontend UI, ties into account creation	2.5 hours	Jack
3	Update server logic for account management; cannot successfully create account without accepting policy	1.5 hours	Pratik
4	Test: while creating an account, don't accept and wait for error, then do accept and continue to onboarding	2 hours	Roan

Acceptance Criteria:

- Users can open and view the entire Terms and Conditions policy
- Users will accept the Terms and Conditions after reading, at the very end of the document
- Users will have to e-sign and abide by the policy to participate in the service
- Users move onto the next step in account creation, onboarding

User Story 21

Old: As a user, I would like to get notifications (through email) for any updates

New: As a user, I would like to be able to set up app notifications for workouts, meals, etc.

Number	Description	Estimated Time	Member Assigned
1	Design and implement a UI in the settings where users can enable notifications	1 Hour	Artem
2	Implement logic to schedule notifications based on the user's selected times for workouts and meals using the device notification system (Expo Notifications)	4 Hours	Artem
3	Test notification functionality to ensure reminders trigger correctly at scheduled times	1 Hour	Artem

Acceptance Criteria:

- Users can enable/disable notifications in the settings
- Given notifications enabled, when it is time for a scheduled workout/meal, a notification will arrive
- Given notifications are disabled, when the scheduled time arrives, a notification will not be sent

Note: We will be using Expo notifications to present the proper functionality, as they are able to be created and shown in the emulation.

Remaining Backlog

Functional Requirements:

General

- ~~As a user, I will partake in an onboarding process that will consist of age, height, weight, goals, and experience~~
- ~~As a user, I would like to be able to create and manage an account for the app, such as creating and changing a username or bio.~~
- ~~As a user, I would like to be able to reset my password.~~
- ~~As a user, I would like to be able to easily navigate to the workout, diet, and social tabs through the use of a menu.~~
- As a user, I would like to be able to read and agree to the Terms and Conditions.
- ~~As a user, I would like to be able to delete my account.~~
- As a user, I would like to be able to set up notifications for workouts, meals, etc.
- ~~As a user, I would like to be able to enable certain accessibility features (dark mode, large text).~~
- ~~As a user, I would like to be able to maintain a week-by-week workout.~~
- As a user, I would like to be able to maintain diet streaks.
- As a user, I would like to be easily able to convert recipe and weight amounts between metric and imperial.
- ~~As a user, I would like to be able to select a location, so I can see gyms and people near me.~~
- As a user, I will be able to bypass manually inputting username and password during login with a Remember Me feature
- As a user, I would like to be able to select a location, so I can see gyms and people near me.
- As a user, I would like to be able to enable certain accessibility features (dark mode, large text).
- As a user, I will be able to export my profile to show a recap of long-term progress

Workout

- ~~As a gym user, I would like to be able to track my workouts by logging the workout, reps, weight, and sets.~~
- ~~As a gym user, I would like to be able to have cardio machines suggested to me for targeted fat loss.~~
- ~~As a gym user, I would be able to upload and save my prior gym routines.~~
- As a gym user, I would like to be able to upload my workouts to a community space.

- ~~As a gym user, I would like to be able to add tags to my workouts to make them easier to find.~~
- ~~As a gym user, I would like to be able to search for different workouts based on the tags assigned to them.~~
- As a gym user, I would like to be able to view shared workouts and do them myself.
- ~~As a gym user, I would like to be able to schedule my workouts on a calendar.~~
- As a beginner on my fitness journey, I would like to know where to start and how to make consistent progress.
- As an experienced athlete, I would be able to identify key techniques and strategies to improve that I otherwise wouldn't have noticed.

Diet

- As a dieter, I would be able to view the macros of different meals from national restaurants.
 - This can be provided through a large database such as [MenuStat](#).
- As a dieter, I would like to be able to chart my daily macro and water intake, and I would like to be able to set macro and water goals to hit daily.
- As a dieter, I would like to be able to set certain junk foods to avoid.
- ~~As a dieter, I will press a button to recalibrate my caloric goals based on recorded behavior.~~
- As a dieter, I will press a button to recalibrate my caloric goals based on recorded behavior.
- As a new user, I will have my day-one caloric goals displayed based on onboarding responses.
- ~~As a dieter, I would like to be able to apply certain tags to my meals to make them easier to find.~~
- ~~As a dieter, I would like to be able to search for different meals using the tags assigned to them.~~
- As a dieter, I would like to filter community meals by servings, calories, protein, cuisine, complexity, and other important components.
- As a user, I would be able to assign macros and ingredients to my meals when I add them.
- As a dieter, I would like to be able to react to different meals using a like feature, emojis, or comments.
- As a dieter, I would like to select ingredients I currently have to create meals from them.
- As a dieter, I would like help settling into a diet with less strict requirements initially, or a cheat day to allow myself an easier transition.
- As a dieter, I would like to know what national chain meals fit into my diet.

Social

- As a dieter, I would like to be able to upload custom meals to a community space.

- As a dieter, I would like to be able to save meals from the community space.
- As a social user, I would like to find other weightlifters nearby.
- As a social user, I would like to be able to “friend” other users.
- As a social user, I would like to be able to message other users in the app.
- As a social user, I would like to view my friends' progress in the app.
- As a social user, I would like to be able to “unfriend” any users.
- As a social user, I would like to be able to block any users who harass me.
- As a social user, I would like to be able to report any users who violate the app TOC.
- As a social user, I would like to be able to share or show off my progress to my friends.
- As a social user, I would like to be able to decide whether my progress is public or available only to specific users.
- As a social user, I would like to be able to view my friends' streaks.
- As a social user, I would like to be able to “like”, react with emojis, and comment on different shared workouts and meals.
- As a social user, I would like to be able to set collaborative goals with my friends, where we all work toward similar goals and hold each other accountable.

AI

- As a user, I would like to be able to have weekly and monthly diet and workout reports generated.
- As a gym user, I would like to be able to ask questions to get clarifications on specifics, such as the best number of reps or sets for a certain exercise.
- As a gym user, I would like to be able to have alternative options suggested if a machine or exercise isn't available or is undesirable.
- As a gym user, I would like to have workouts change based on the inability to do certain workouts, such as due to injury.
- As a gym user, I would like to achieve a natural weight progression based on how often I work out on a specific exercise.
- As a gym user, I would like advice on how to recover from an injury.
- As a gym user, I would be able to take a video of my lifts and harness AI to correct my form if need be.
 - Due to the large scale of this story, this will take significantly longer than other stories, approximately 30 hours. This was discussed with the coordinator.
 - The primary challenges in this story come from being able to analyze form using AI, as it would require a large dataset of exercise videos to train on to compare an arbitrary video. It would not only have to recognize the technique but compare it to a “perfect” technique as well, and pass the results to an LLM to formulate feedback.

- As a gym user, I would like exercises recommended to me based on my experience level, which I have stated in my onboarding.
- As a gym user, I would like to be able to generate a gym routine based on the muscle group to train.
- As a dieter, I would be able to have recipes suggested to me for various goals related to weight loss or muscle gain.
- As a beginner in my fitness journey, I would like tips or advice on how to do a specific exercise for the first time, embedded as a help button.

Additional Stories: